

IAM in GCP

Identity and Access Management (IAM) in Google Cloud Platform (GCP)

Overview

Identity and Access Management (IAM) in Google Cloud Platform (GCP) is a framework that enables you to manage **who (identity)** has **what access (roles)** to **which resources**. IAM allows organizations to implement and enforce fine-grained access controls and ensure that the right individuals and services have the appropriate permissions.

IAM is foundational to GCP's security model, offering a unified system for managing access across all GCP services.

Core Concepts

1. Identities

- Represents who is requesting access.
- Types of identities:
 - Google Account: Individual user accounts.
 - Service Account: Special accounts for applications or virtual machines.
 - Google Groups: Collection of Google accounts.
 - Cloud Identity or G Suite domains: Organizations using Google Workspace.
 - External identities via Identity Federation.

2. Resources

- The target that the identity is trying to access.
- Examples: Compute Engine VM instances, Cloud Storage buckets, BigQuery datasets.

3. Roles

- Define what permissions an identity has.
- Types of roles:
 - **Basic roles:** Primitive roles such as Owner, Editor, and Viewer.
 - **Predefined roles:** Granular roles curated by Google (e.g., BigQuery Data Viewer).
 - **Custom roles:** Roles created by users for specific permissions.

4. Policies

- Bindings that connect identities with roles to resources.

- IAM policies are hierarchical and can be applied at the organization, folder, project, or resource level.

Use Cases

1. Least Privilege Access

- Ensures users and services have only the permissions they need, and no more.

2. Multi-tenancy and Project Isolation

- Enables organizations to manage multiple teams and projects securely under a single GCP organization.

3. Audit and Compliance

- IAM policies and logs support regulatory compliance through visibility and control.

4. Automated Workflows

- Service accounts can be used by applications and automation scripts to access resources securely.

5. Secure API Access

- Manage which identities (users or services) can interact with GCP APIs.

6. Delegated Administration

- Teams or departments can manage their resources without needing full administrative privileges.

Benefits of IAM in GCP

1. Centralized Access Management

- Unified system to manage permissions across all GCP services.

2. Fine-Grained Control

- Enables detailed permission management with predefined and custom roles.

3. Improved Security Posture

- Enforces the principle of least privilege, reducing the risk of unauthorized access.

4. Auditability and Transparency

- Integration with Cloud Audit Logs for tracking changes and access attempts.

5. Scalability

- Suitable for organizations of all sizes, from startups to large enterprises with complex hierarchies.

6. Flexibility

- Supports cross-project access and hybrid identity scenarios (e.g., on-premises and cloud integration).

7. Integration with Other GCP Services

- IAM is tightly integrated with services like Cloud Storage, BigQuery, Pub/Sub, and more, streamlining access control configuration.

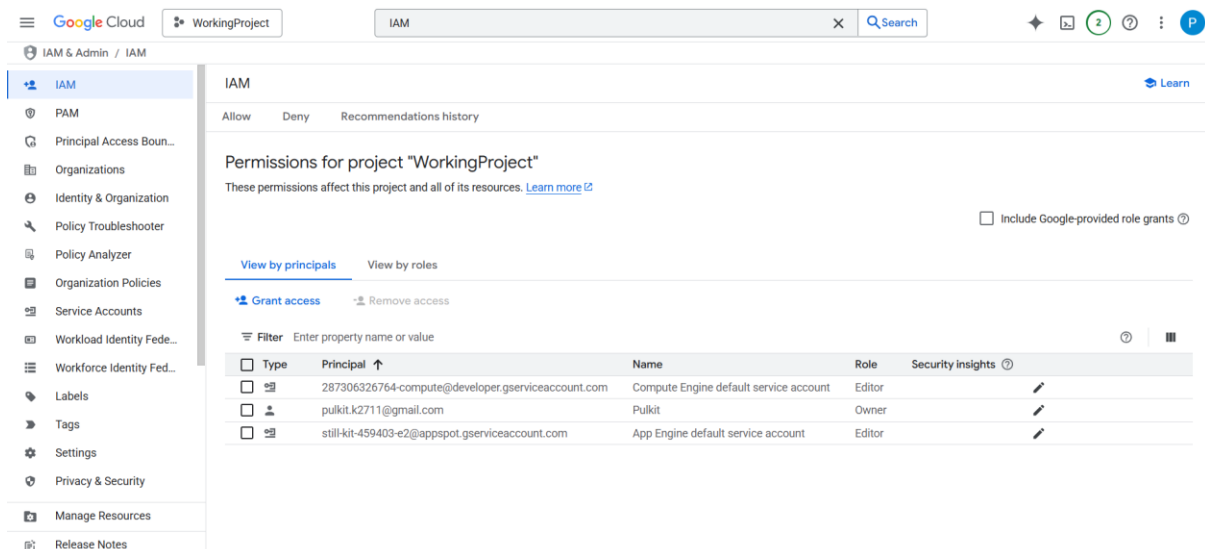
Best Practices

- Use **predefined roles** whenever possible.
- Follow the **principle of least privilege**.
- Regularly **audit IAM policies**.
- Use **service accounts** for applications instead of user credentials.
- Use **organization policies** to set governance rules at scale.
- Enable **Cloud Audit Logs** for visibility into access patterns.

IAM in GCP is a powerful tool that helps organizations manage access securely and efficiently. By using IAM effectively, you can ensure that your cloud resources are protected against unauthorized access while maintaining operational flexibility.

To begin with the Lab

1. From your GCP console, search and navigate to IAM. Here you will see that there are three principal accounts.
2. First is your main account or root account, which you are using to work on GCP, and the others are for Compute Engine and App Engine.



The screenshot shows the Google Cloud IAM console interface. The left sidebar contains navigation links for IAM, PAM, Principal Access Boundaries, Organizations, Identity & Organization, Policy Troubleshooter, Policy Analyzer, Organization Policies, Service Accounts, Workload Identity Federation, Workforce Identity Federation, Labels, Tags, Settings, Privacy & Security, Manage Resources, and Release Notes. The main content area displays the 'Permissions for project "WorkingProject"' page. It includes a search bar, a 'Learn' link, and a table of permissions. The table has columns for Type, Principal, Name, Role, and Security insights. There are three entries in the table:

Type	Principal	Name	Role	Security insights
Service account	287306326764-compute@developer.gserviceaccount.com	Compute Engine default service account	Editor	Security insights
User	pulkit.k2711@gmail.com	Pulkit	Owner	Security insights
Service account	still-kit-459403-e2@appspot.gserviceaccount.com	App Engine default service account	Editor	Security insights

3. Now, to better understand roles and access, we are going to create a Virtual Machine Instance.
4. For that, navigate to EC2 and create a VM. Here you can see that our VM is up and ready.

VM instances

Filter Enter property name or value

☐ Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	iam-vm	us-central1-f			10.128.0.16 (nic0)	34.58.169.12 (nic0)	SSH ▾

5. Now, come back to IAM and from the IAM dashboard page, click on Grant Access.

IAM

[Learn](#)

Allow Deny Recommendations history

Permissions for project "WorkingProject"

These permissions affect this project and all of its resources. [Learn more](#)

☐ Include Google-provided role grants

View by principals View by roles

Grant access Remove access

Filter Enter property name or value

☐ Type	Principal ↑	Name	Role	Security insights ⓘ
☐	287306326764-compute@developer.gserviceaccount.com	Compute Engine default service account	Editor	
☐	pulkit.k2711@gmail.com	Pulkit	Owner	
☐	still-kit-459403-e2@appspot.gserviceaccount.com	App Engine default service account	Editor	

6. In the add principals' area, you need to put an email address to which you want to give access and then you need to assign Compute Viewer role to it. Click on Save.

Resource

WorkingProject

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals *

@gmail.com

X

?

Assign roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *

Compute Viewer

▼

Read-only access to get and list information about all Compute Engine resources, including instances, disks, and firewalls. Allows getting and listing information about disks, images, and snapshots, but does not allow reading the data stored on them.

IAM condition (optional) ?

+ Add IAM condition

+ Add another role

Save

Cancel

7. In the permission for project area, you can see that our new email address has been added and it has a role attached to it.
8. Also, you can notice that we have given access to the user only to the Working Project.

Permissions for project "WorkingProject"

These permissions affect this project and all of its resources. [Learn more](#)

☐ Include Google-provided role grants

[View by principals](#)
[View by roles](#)
[Grant access](#)
[Remove access](#)
 Filter Enter property name or value

☐ Type	Principal ↑	Name	Role	Security insights
☐	287306326764-compute@developer.gserviceaccount.com	Compute Engine default service account	Editor	?
☐	@gmail.com		Compute Viewer	?
☐	pulkit.k2711@gmail.com	Pulkit	Owner	?
☐	still-kit-459403-e2@appspot.gserviceaccount.com	App Engine default service account	Editor	?

9. If you go to any other project, you will notice that the user and role you added do not exist here.

10. Because the user and role you grant access to exist in another project, not this one.

Permissions for project "Rb1205"

These permissions affect this project and all of its resources. [Learn more](#)

☐ Include Google-provided role grants

[View by principals](#)
[View by roles](#)
[Grant access](#)
[Remove access](#)
 Filter Enter property name or value

☐ Type	Principal ↑	Name	Role	Security insights
☐	613113882640-compute@developer.gserviceaccount.com	Compute Engine default service account	Editor	?
☐	first-account@rb1205.iam.gserviceaccount.com	first-account	Cloud Infrastructure Manager Agent Compute Admin Service Account User	?
☐	pulkit.k2711@gmail.com	Pulkit	Owner	?
☐	rb1205@appspot.gserviceaccount.com	App Engine default service account	Editor	?

11. Now, in another browser, we are going to log in to Google Cloud using the same email we have given access to.

12. In another account, we have opened the Google Cloud Console, and we can see that we can view our instances.

13. But if you try to do something with it, then it will deny your access because we have applied only the view role to it.

Google Cloud WorkingProject Compute

Compute Engine

Overview

Security risk overview

Virtual machines

Migrate to Virtual Mach...

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Committed use discou...

Reservations

Storage

Disks

Storage Snapshots

Marketplace

Release Notes

VM instances

Create Instance Import VM Refresh

Instances Observability Instance schedules

1 Start / Resume Stop Suspend Reset Create a group based on this vm Run maintenance More actions

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	iam-vm	us-central1-f			10.128.0.16 (nic0)	34.58.169.12 (nic0)	SSH

Related actions

- Explore protection summary: Identify gaps in data protection at no cost and configure VM backups
- Monitor VMs: View outlier VMs across metrics like CPU and network
- Explore VM logs: View, search, analyze, and download VM instance logs
- Set up firewall rules: Control traffic to and from a VM instance
- Patch management: Schedule patch updates and view patch compliance on VM instances
- Load balance between VMs: Set up Load Balancing for your applications as your traffic and users grow

14. Now, if you try to access any service other than compute, then you will get access denied.

15. So, we tried to open the SQL database service, and we got the message that we'll be needing additional access.

Google Cloud WorkingProject sql

Compute Engine

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Machine images

TPUs

Committed use discou...

You need additional access

You need additional access to the project: WorkingProject

This could be because you have insufficient permissions to access the resource, or because a Principal Access Boundary policy is blocking your access to the resource.

To request access, contact your project administrator and provide them a copy of the following information:

Troubleshooting info:

Principal: lordstar697@gmail.com

Resource: still-kit-459403-e2

Troubleshooting URL: console.cloud.google.com/iam-admin/troubleshooter;permissions=cloudsql.instance

Missing permissions:

cloudsql.instances.list

If your administrator is unable to help, then [contact support](#).

16. Based on the requirements, you can now give access to the user to the services of your choice.

17. Also, you can explore the other options like roles, security, and privacy, etc.